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ORIGINAL ARTICLE

Plasma choline, homocysteine and vitamin status in healthy adults supplemented with krill oil: a pilot study

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ABSTRACT

Plasma concentrations of metabolites along the choline oxidation and tryptophan degradation pathways have been linked to lifestyle diseases and dietary habits. This study aimed to investigate how krill oil, a source of ω-3 polyunsaturated fatty acids (PUFAs) with a high phosphatidylcholine content, affected these parameters. The pilot study was conducted as a 28 days intervention in 17 healthy volunteers (18–36 years), who received a supplement of 4.5 g krill oil per day, providing 833 mg ω-3 PUFAs, and 1750 mg phosphatidylcholine. Krill oil supplementation increased fasting plasma choline (+28.4%, $p < .001$), betaine (+26.6%, $p < .001$), dimethylglycine (+33.7%, $p < .001$) and sarcosine (+16.8%, $p < .001$), whereas no statistically significant changes were seen for plasma glycine, serine, methionine, total homocysteine, cysteine, cystathionine, methionine sulfoxide, folate, cobalamin, B_2-, B_3-, and B_6 vitamers, tryptophan, kynurenines, nicotinamide, vitamin A and vitamin E. In summary, krill oil supplementation influenced choline metabolite levels, but not plasma metabolites of the tryptophan-kynurenine-nicotinamide pathways and vitamins. These observations should be confirmed in a placebo-controlled trial, including an ω-3 PUFA supplement without phospholipids to explore the potential additive effects of the different active ingredients.

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Introduction

Elevated plasma total homocysteine (tHcy) is related to increased risk of atherosclerosis and cardiovascular disase (CVD) [1], but lowering of tHcy with B-vitamins has not improved prognosis among CVD patients [2]. Hcy may be remethylated back to methionine (Met) through the B_{12}-dependent methionine synthase, using 5′-methyltetrahydrofolate (mTHF) as the methyl donor. In the liver and kidney, Hcy may also be remethylated by betaine-homocysteine methyltransferase (BHMT), using betaine as the methyl donor. This couples the Hcy metabolism to the choline oxidation pathway, and the reaction yields dimethylglycine (DMG). Recently, it has been found that higher plasma concentration of DMG is associated with increased CVD risk [3,4]. Irreversible catabolism of Hcy takes place through the B_6-dependent transsulfuration pathway (Figure 1).

Previous findings suggest that in patients with suspected angina pectoris, elevated levels of plasma kynurenines predicted increased risk of acute myocardial infarction [5], but it is still unknown whether dietary interventions that reduce plasma kynurenine levels will reduce risk of coronary artery disease. Kynurenines are a collective term of the compounds originating from the catabolism of the amino acid tryptophan (Trp) through the kynurenine pathway [6], which depends on vitamins B_2 and B_6 as cofactors [7,8]. One of the kynurenines, 3-hydroxyanthranilic acid, is further metabolized to quinolonic acid and may ultimately form vitamin B_3 [9]. This metabolic pathway is referred to as the Trp-kynurenine-nicotinamide (NAM) pathway.

ω-3 PUFA supplementation may decrease plasma tHcy in patients diagnosed with diabetic dyslipidemia [10], myocardial infarction [11] and hyperlipidemia [12], but the effect of this tHcy lowering on disease outcome is not known. Studies have shown that phospholipids may improve bioavailability of ω-3 PUFAs [13,14]. Moreover, it has been reported that a phospholipid–protein complex from krill reduced plasma tHcy but increased the concentration of DMG in rats [15]. The mechanism behind plasma tHcy-lowering by ω-3 PUFA is unknown, but long-term ω-3 supplementation was shown to prevent age-related decrease in the expression of cystathionine β-synthase (CBS) and BHMT, enzymes involved in Hcy metabolism in mice, indicating that ω-3 PUFA can increase both remethylation and transsulfuration of Hcy [16]. In addition, the high amount of phosphatidylcholine (PC) in krill oil makes it a good source of choline, which is an essential nutrient and a major source of dietary methyl groups [17]. Choline deficiency may lead to reduced capacity for Hcy methylation, thus elevating plasma Hcy levels, although this has only been

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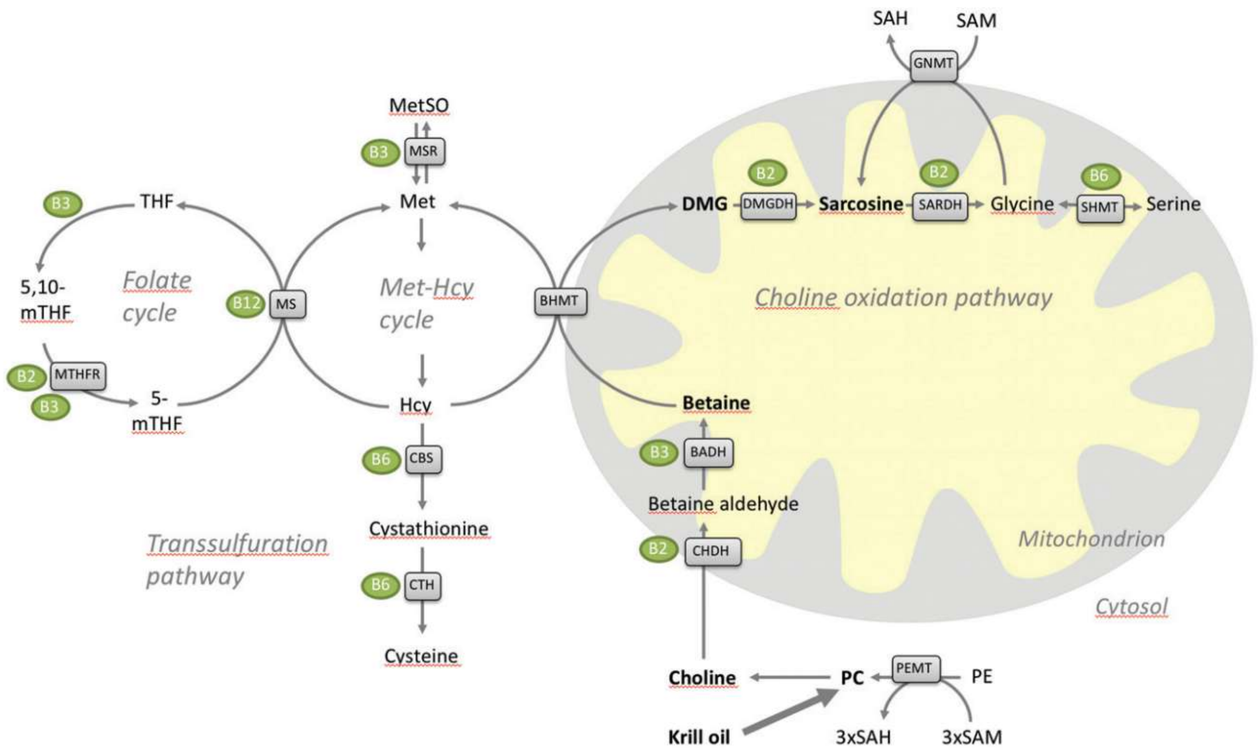

Figure 1. Overview of the folate cycle, homocysteine metabolism and the choline oxidation pathway. Adapted from Ref. [38]. BADH: betaine aldehyde dehydrogenase; BHMT: betaine homocysteine methyltransferase; CBS: cystathionine beta-synthase; CHDH: choline dehydrogenase; CTH: cystathionine gamma lyase; DMG: dimethylglycine; DMGDH: dimethylglycine dehydrogenase; Hcy: homocysteine; Met: methionine; MS (MTR): methionine synthase; PC: phosphatidylcholine; PE: phosphatidylethanolamine; PEMT: phosphatidylethanolamine methyltransferase; SAH: S-adenosylhomocysteine; SAM: S-adenosylmethionine; SARDH: sarcosine dehydrogenase; SHMT: cytosolic (SHMT1) or mitochondrial (SHMT2) serine hydroxymethyltransferase; THF: tetrahydrofolate.

demonstrated at very high levels of Met supplementation [18]. In addition, choline is needed for phospholipid and neurotransmitter synthesis [17].

Although fish intake is not an important determinant of the metabolites in the kynurenine pathway, an inverse association between dietary ω-3 PUFA and the kynurenine-to-tryptophan ratio (KTR) and neopterin was demonstrated in patients with coronary artery disease, indicating that diet can influence these parameters [19].

In the present study, we explored the potential effect of krill oil (Rimfrost Sublime™, Rimfrost AS, Fosnavag, Norway) on plasma concentrations of components of the choline oxidation and the Trp-Kynurenine-NAM pathways as well as vitamin markers.

Material and methods

Study design

In this pilot study, all participants received the supplement for 28 days. The study protocol was approved by the Regional Ethics Committee of Northern Norway (registration no. 2013/2152). All procedures were carried out in accordance with the ethical standards of the Declaration of Helsinki of 1975, revised in 2008, for experiments involving humans. Eighteen subjects (15 female, 3 male) aged 18–36 years, with a body mass index (BMI) of 17–29.9 in kg/m^2 and in a generally good health, were recruited through advertisements presented at the University of Bergen and the Bergen University College (Norway). Exclusion criteria were diabetes type 1 or 2, hyperlipidemias, cardiovascular pathologies, gastrointestinal disorders, acute or chronic inflammatory conditions, liver dysfunctions, pregnancy (all self-reported), obesity (defined as a BMI ≥ 30), inability to swallow capsules and unwillingness to take blood samples. After written informed consent was provided, blood samples were drawn, and body weight and height were recorded to calculate BMI. Study participants were interviewed about their dietary intake of foodstuffs rich in ω-3 fatty acids and PLs, focusing on the amount and frequency of the last month intake. Participants were advised to maintain a steady intake of these food items during the study period, refrain from the use of all nutritional supplements, and to maintain their body weight. At the end of the study, all participants were asked to answer questionnaires to assess self-reported compliance and product tolerance.

All 18 subjects completed the intervention period. However, one subject (male) was excluded from the final analysis due to lack of compliance with the protocol as revealed by the questionnaires. For the 17 individuals included in the final analysis, the mean age was 23 years (range 20–33). All examinations and collection of blood samples were performed at the Department of Clinical Science, University of Bergen, Norway. Primary outcome measures, serum fatty acid composition and blood lipid parameters, were published in Berge et al. [20], while secondary outcomes measures, choline and its metabolites, tryptophan metabolites, and vitamins in plasma, are reported here.

Intervention

The intervention used was krill oil capsules (RIMFROST Sublime®, Rimfrost AS). The lipid- and fatty acid composition of the oil is previously presented [20]. Each capsule (500 mg oil) contained 210 mg PLs, of which 92.5% was phosphatidylcholine, and 120 mg ω-3 PUFA, of which 60 mg was eicosapentaneoic acid (C20:5ω-3, EPA) and 32.5 mg docosahexaenoic acid (C20:6ω-3, DHA). The participants received nine capsules per day for 28 days, corresponding to a daily dose of 833 mg EPA and DHA, and 1750 mg PC. Moreover, this dose provided 864 µg astaxanthin per day.

Blood sample collection and biochemical analyses

Blood samples were collected from subjects fasted overnight and seated for 15 min, at the beginning (baseline) and end of the study. Blood samples were centrifuged and EDTA-plasma was collected after a minimum of 15 min and maximum of 30 min at room temperature while serum was collected after 30 min at room temperature. All samples were aliquoted and stored at −80 °C for further analysis. The plasma metabolites, kynurenines, amino acids, B-vitamin and lipid soluble vitamins were measured by liquid or gas chromatography/tandem-mass spectrometry [21,22] at the laboratory of Bevital AS (www.bevital.no). Lipids were measured enzymatically in fasting serum samples on a Hitachi 917 system (Roche Diagnostics GmbH, Mannheim, Germany) using the triacylglycerol (TAG; GPO-PAP), cholesterol (CHOD-PAP), HDL-cholesterol plus and LDL-cholesterol plus kit from Roche Diagnostics, the non-esterified fatty acid (NEFA FS) kit and the Phospholipids FS kit from Diagnostic Systems GmbH (Holzheim, Germany).

Statistics

Plasma lipids are reported as geometric means and their geometric standard deviations. Statistical difference between baseline and end of study was analyzed using paired *t*-tests on log-transformed values. $p < .05$ were considered significant.

Metabolite concentrations are reported as geometric means and their geometric standard deviations. Paired *t*-tests were performed to assess differences before and after supplementation, and the standardized mean changes (SMC) were calculated based on the log-transformed metabolite concentrations. Corrections for multiple comparisons were performed according to Benjamini and Hochberg (1995) [23]. $p < .05$ were considered significant. R version 3.3.1 (The R Foundation for Statistical Computing, Vienna, Austria) was used for statistical analyses, and SMC was calculated with the *effsize* package, while correction for multiple comparison was performed with the function p.adjust from the package Stat, which provides *p* values adjusted for the critical value.

Table 1. Study population and fasting lipid parameters at baseline and after 4 weeks krill oil supplementation.

Parameter	Baseline	End
Sex		
Male, *n*	2	–
Female, *n*	15	–
Age[a], years (range)	23 (20–33)	–
Body mass index[b], kg/m^2	20.9 (1.11)	–
Cholesterol[b], mmol/L	4.35 (1.14)	4.58 (1.15)**
Phospholipids[b], mmol/L	2.42 (1.15)	2.54 (1.16)**
Non-esterified fatty acids[a], mmol/L	0.42 (1.77)	0.33 (1.69)
TAG[b], mmol/L	0.76 (1.42)	0.69 (1.53)*
HDL-cholesterol/LDL-cholesterol ratio[b]	0.65 (1.25)	0.67 (1.35)
TAG/HDL-cholesterol ratio[b]	0.47 (1.47)	0.40 (1.75)**

[a]Values are mean with range ($n = 17$).
[b]Values are geometric means with geometric standard deviations ($n = 17$). Statistical difference between baseline and end of study was determined using paired *t*-tests (*$p < .05$, **$p < .01$).
HDL-cholesterol: high-density lipoprotein cholesterol; LDL-cholesterol: low-density lipoprotein cholesterol; TAG: triacylglycerol.

Results and discussion

Baseline characteristics and fasting serum lipids

In this pilot study, supplementation with krill oil for 28 days in young, healthy adults was previously reported to result in a significantly reduced large VLDL and chylomicron particle concentrations [20]. Krill oil intake was associated with lower fasting plasma TAG concentration and TAG/HDL-cholesterol ratio at the end of the study (Table 1). Higher levels of plasma phospholipids and cholesterol was observed at the end of the intervention, while non-esterified fatty acids levels were lowered without reaching statistical significance ($p = .052$).

The observed TAG lowering is in line with the results from a recent meta-analysis on randomized controlled trials with krill oil in humans [24], which in addition found plasma LDL-cholesterol to be significantly reduced. The findings support that neutroceutical and functional food ingredients can be important additions to conventional medication in persons with dyslipidaemia [25]. Krill oil has many overlapping effects with fish oil, due to their similarity in EPA and DHA content, but a lower dose of EPA and DHA from krill oil may be needed to get a biological effect due to differences in bioavailability of phospholipids from krill oil and TAG from fish oil [26]. Also, the high amount of PC in krill oil may result in additional health benefits by providing the essential nutrient choline [27]. In line with this, plasma phospholipids were higher after the krill oil intervention. Further, we investigated the influence of the krill oil supplement on plasma metabolites that could be influenced by choline and/or EPA and DHA.

Plasma one-carbon and choline oxidation metabolites

Metabolite concentrations at baseline and after the intervention, as well as the SMC, are summarized in Figure 2. Krill oil supplementation for 28 days was associated with increased plasma concentrations of choline (SMC = 0.72, $p < .001$), betaine (SMC = 0.40, $p < .001$), DMG (SMC = 0.50, $p < .001$) and sarcosine (SMC = 0.59, $p < .001$), but no associations were observed with plasma glycine or

	Baseline	End	Standardized mean change	sig.
One-carbon metabolism				
Methionine, µmol/L	24.6 (1.18)	26.1 (1.15)		0.168
Homocysteine, µmol/L	8.49 (1.15)	8.27 (1.16)		0.455
Cystathionine, µmol/L	0.18 (1.46)	0.19 (1.47)		0.862
Cysteine, µmol/L	261 (1.10)	261 (1.09)		0.988
Methionine sulfoxide, µmol/L	0.72 (1.26)	0.75 (1.21)		0.644
Choline oxidation pathway				
Choline, µmol/L	7.78 (1.19)	9.99 (1.20)		<0.001
Betaine, µmol/L	25.2 (1.45)	31.7 (1.42)		<0.001
DMG, µmol/L	4.75 (1.37)	6.35 (1.48)		<0.001
Sarcosine, µmol/L	2.56 (1.19)	2.99 (1.14)		<0.001
Glycine, µmol/L	205 (1.38)	202 (1.35)		0.862
Serine, µmol/L	117 (1.18)	117 (1.21)		0.965
Tryptophan and kynurenines				
Tryptophan, µmol/L	61.2 (1.19)	65.3 (1.17)		0.357
Kynurenine, µmol/L	1.29 (1.11)	1.36 (1.11)		0.248
Kynurenine-Tryptophan ratio	21.1 (1.15)	20.9 (1.13)		0.862
Hydroxykynurenine, nmol/L	34.3 (1.22)	36.5 (1.24)		0.387
Kynurenic acid, nmol/L	41.7 (1.42)	42.1 (1.39)		0.965
Xanthurenic acid, nmol/L	14.4 (1.59)	16.1 (1.56)		0.455
Anthranilic acid, nmol/L	14.1 (1.3)	13.8 (1.38)		0.862
Hydroxyanthranilic acid, nmol/L	29.7 (1.39)	30.6 (1.31)		0.862
Quinolinic acid, nmol/L	343 (1.19)	341 (1.19)		0.921
Neopterin, nmol/L	9.26 (1.33)	9.88 (1.36)		0.353
Water soluble vitamins				
Riboflavin, nmol/L	10.3 (1.87)	11.2 (1.80)		0.455
Flavin mononucleotide, nmol/L	15.3 (1.40)	10.4 (5.08)		0.455
Nicotinamide, nmol/L	366 (1.32)	377 (1.32)		0.862
Methylnicotinamide, nmol/L	124 (1.85)	163 (1.62)		0.156
Pyridoxal, nmol/L	11.9 (1.37)	12.6 (1.51)		0.357
Pyridoxalphosphate, nmol/L	71.1 (1.54)	81.3 (1.65)		0.122
Pyridoxic acid, nmol/L	29.3 (1.41)	32.4 (1.50)		0.327
Methyltetrahydrofolate, nmol/L	14.5 (1.51)	13.6 (1.46)		0.387
Cobalamin, pmol/L	427 (1.43)	409 (1.35)		0.455
Methylmalonic acid, µmol/L	0.16 (1.44)	0.16 (1.42)		0.965
Fat soluble vitamins				
Vitamin A, µmol/L	2.04 (1.24)	2.21 (1.28)		0.077
α-Tocopherol, µmol/L	26.5 (1.20)	28.3 (1.23)		0.082
γ-Tocopherol, µmol/L	1.34 (1.29)	1.55 (1.38)		0.120
Vitamin D3, nmol/L	56.5 (1.57)	57.3 (1.48)		0.914

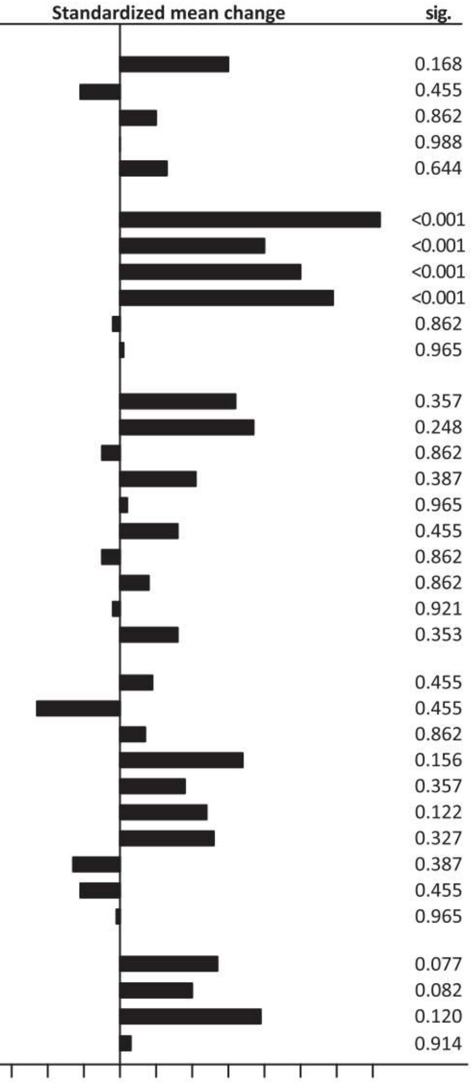

Figure 2. Plasma concentration of metabolites at baseline and after 4 weeks krill oil supplementation. Values are geometric means and geometric standard deviations, and were compared by paired *t*-tests and adjusted for multiple comparisons using the method of Benjamini and Hochberg. The plot indicates the standardized mean change, calculated on the log-transformed metabolite concentrations.

serine. The krill oil supplement used in this study contributes with almost 2 g/day of phosphatidylcholine. As the daily intake of phospholipids, of which the majority is comprised of phosphatidylcholine, is estimated to be 2–8 g [28], this is a considerable amount. Our findings suggest that intake of this dose of krill oil is sufficient to affect metabolites along the choline oxidation pathway. There were no associations for Met, tHcy, Met sulfoxide, or the transsulfuration metabolites cystathionine or cysteine. Thus, the increased choline and betaine level, potentially providing an increased availability of methyl donors, did not significantly influence the metabolites of the Met-Hcy cycle or transsulfuration, although Met-levels tended to increase (p=.17). Altogether, our data suggest that an increased dietary intake of choline can increase BHMT-mediated Hcy remethylation, thereby increasing DMG- and sarcosine levels. This is in line with results from long-term ω-3 PUFA supplementation in mice, showing that age-related decline in BHMT as well as CBS is prevented by ω-3 PUFA [16]. Sarcosine is metabolized to glycine and further to serine [29], but plasma glycine and serine did not respond to krill oil supplementation. This may be explained by glycine and serine being involved in numerous additional pathways and their plasma concentrations are under strict metabolic control [29]. An extended krill oil intervention study beyond 28 days should be performed to determine if the increased plasma choline level could lower plasma Hcy levels by reducing the need for *de novo* choline synthesis [30]. Low tHcy may have beneficial effects during pregnancy and reduce risk of cardiovascular disease [1], cancer [31], cognitive decline [32], and bone fractures [33].

The enzymes of the choline oxidation pathway are potentially regulated by altered levels of vitamin co-factors, including vitamins B_2, B_3 and B_6, but none of the water-soluble or fat-soluble vitamins analyzed were significantly affected by krill oil.

Tryptophan and kynurenines

We did not observe any differences in the plasma concentrations of metabolites of the Trp-Kynurenine pathway, consistent with the recent report from Karlsson et al. [19] showing that fish and ω-3 PUFAs was not an important determinant of individual metabolites in the kynurenine pathway. In the Nurses' Health Study, a high choline intake was associated with an improved plasma level of inflammatory biomarkers, such as adiponectin, resistin and C-reactive protein. The plasma levels of kynurenine, 3-hydroxykynurenine, 3-hydroxyanthranilic acid and kynurenic acid have been reported to be associated with chronic and acute inflammation [5,34–36], and a high kynurenine:Trp ratio can predict acute coronary events in older adults [37]. It is possible that these parameters could have been influenced by an extended krill oil intervention or in populations with inflammatory diseases.

Strengths and limitations

The main limitation of this study is the lack of a control group, which limits the possibility to attribute the observed changes to the krill oil supplement. Additionally, the small sample size is an important limitation, and may have prevented the detection of additional clinically relevant effects. As the requited population were predominantly female, gender-specific effects cannot be excluded. A strength of the study was a high compliance rate (>99%), and that all included participants completed the study.

Krill oil contains omega-3 fatty acids as well as PCs, of which both may be related to the observed changes in plasma metabolites. The presence of phospholipids may potentially influence the effect of the ω-3 PUFAs and vice versa.

Conclusions

In this pilot study, a phospholipid enriched ω-3 PUFA supplement was associated with increased plasma concentrations of choline and metabolites along the choline oxidation pathway. This indicates that dietary choline in the form of krill oil could potentially increase processes dependent on methyl-donors. The participants were healthy and had a low mean age (23 years). It would be interesting to repeat the study in older populations including patients, such as patients with dyslipidemia. Our observations should be confirmed in a placebo-controlled trial, including an ω-3 PUFA supplement without phospholipids to explore the potential additive effects of the different active ingredients.

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Disclosure statement

Rimfrost AS employed I.B. and M.S.R. at the time of the study. No potential conflict of interest was reported by the authors.

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